

Scenario-Based Questions

For the following questions, consider this database schema with four tables: **EMPLOYEE**, **DEPARTMENT**, **GRADE**, and **PROJECT**.

Table: EMPLOYEE

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450	NULL	10
7788	SCOTT	ANALYST	7566	1987-04-19	3000	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000	NULL	10

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1987-05-23	1100	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300	NULL	10

Table: DEPARTMENT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Table: GRADE (Represents Salary Grade)

GRADE	LOSAL	HISAL
1	700	1200

GRADE	LOSAL	HISAL
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

Section 1: Basic SELECT Queries

Q: Display all information about all employees.

Q: Display the employee number, name, and salary of all employees.

Q: Display the unique job titles from the EMPLOYEE table.

Q: Display the employee name and their annual salary (SAL * 12).

Q: Display the details of all employees from department number 10.

Q: Display the name and job of employees who are 'CLERK'.

Q: Display the name of employees whose name starts with 'S'.

Q: Display the name of employees where the second letter of their name is 'L'.

Q: Display the name of employees who have 'AR' in their name.

Q: Display the details of employees who were hired in the year 1981.

Q: Display the name and salary of employees who earn more than 2500.

Q: Display the details of employees who are not 'MANAGER'.

Q: Display the details of employees who have a commission.

Q: Display the details of employees who do not have a manager.

Q: Display the names of all employees from department 30 who are 'SALESMAN'.

Q: Display the names of all 'CLERK's and 'ANALYST's.

Q: Display the names of employees whose salary is between 1000 and 2000.

Section 2: Ordering and Grouping

Q: Display employee details sorted by their name in alphabetical order.

Q: Display employee details sorted by department number, and then by salary in descending order.

Q: Display the total number of employees.

Q: Find the total salary paid to all employees.

Q: Find the average salary of all employees.

Q: Find the maximum and minimum salary from the EMPLOYEE table.

Q: Count the number of employees in each department.

Q: Find the average salary for each job type.

Q: Find the total salary for each department.

Q: Display the department numbers that have more than 3 employees.

Q: Display the jobs where the average salary is more than 2000.

Section 3: Joins and Subqueries

Q: Display the employee name and their department name.

Q: Display the employee name, department name, and location for all employees.

Q: Display the name of the employee and the name of their manager.

Q: Display the employee name, salary, and salary grade.

Q: Display the employee name for those working in 'NEW YORK'.

Q: Find the name of the department where 'SMITH' works.

Q: Display the names of employees who earn more than 'ALLEN'. (Subquery)

Q: Display the names of employees who work in the same department as 'SCOTT'.

Q: Display the names of employees who have the same job as 'JONES'.

Q: Display the names of managers who manage more than 2 employees.

Section 4: Functions (String, Numeric, Date)

Q: Display the employee names in lowercase.

Q: Display the first three characters of each employee's name.

Q: Display the length of each employee's name.

Q: Display the salary of each employee rounded to the nearest thousand.

Q: Display the remainder of dividing the salary by 1000 for each employee.

Q: Display the current date and time.

Q: Display the year each employee was hired.

Q: Display the name of the month each employee was hired.

Q: Display the number of years each employee has been with the company (assuming the current year is 2025).

Section 5: DDL and DML

Q: Write the SQL command to create the DEPARTMENT table with DEPTNO as the primary key.

Q: Add a new column PINCODE of type INT to the DEPARTMENT table.

Q: Insert a new record for department 50, 'HR', located in 'HOUSTON' into the DEPARTMENT table.

Q: Update the location of the 'SALES' department to 'ATLANTA'.

Q: Delete the 'OPERATIONS' department from the table.